

Kunal Sehta

Technical Specialist: AI

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Innovate as an AI Specialist, leveraging five-plus years of rich experience in Generative AI, Natural Language Processing, Deep Learning, and Machine Learning, while focusing on creating transformative AI solutions that cater to various sectors.



PROFILE SUMMARY

- ❖ Technology-driven professional with experience of **5+ years in Artificial Intelligence**, with a focus on **Natural Language Processing and Machine Learning**, showcasing a dedication to fostering innovation and enhancing operational efficiency.
- ❖ Led the development of advanced **AI applications at AIT Global**, integrating the latest LLM technologies to enhance user experience and optimize operational efficiency. Implemented **LLM-driven solutions** within applications and leveraged **Docker** for scalable deployment and seamless integration.
- ❖ Cultivated extensive expertise across multiple roles, concentrating on the design and execution of AI-driven solutions, while honing critical skills in model optimization, data analysis, and algorithm development.
- ❖ Engineered a sophisticated **Retrieval-Augmented Generation (RAG) system** that led to a significant **35% boost** in contextual accuracy for chatbot applications, showcasing a robust capability to improve user engagement.
- ❖ Showcased outstanding leadership and teamwork skills by collaborating with diverse teams to implement large language models capable of processing millions of queries each month, ensuring optimal integration and performance.
- ❖ Extensive knowledge of cutting-edge NLP techniques and frameworks such as LangChain and LLaMA Index, facilitating the creation of modular and adaptable AI solutions designed to meet unique business requirements.
- ❖ Played a key role in a project that led to a **20% rise** in customer satisfaction by creating an innovative chatbot, reflecting a dedicated emphasis on user-centered design and functionality.



CORE COMPETENCIES



Scalable AI Solutions Development



Natural Language Processing



Cloud Computing Solutions



Data-Driven Decision Making



LLM base Application Deployment



Advanced Machine Learning



Advance Data Visualization



Algorithm Development and Optimization



User Experience Optimization



TECHNICAL SKILLS

Programming: Python (NLTK, spaCy, Gensim, Hugging Face, Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Keras, Pandas, NumPy, SciPy), R (dplyr, ggplot2, Esquisse, mlr3, Lubridate, caret, Glmnet), Lang Chain, LLAMA Index

Data Analytics: POWER BI, Tableau, MS Excel, Jupiter notebook, PySpark, MongoDB (NoSQL)

AWS: EC2, S3, Athena, Redshift, Glue, Kinesis, Lambda, EMR, ELB, SQS, SNS, KMS, Cloud formation, Event bridge

Data: MYSQL, RDS, Aurora, Postgrese SQL, Hadoop, Spark, Hive, HBase, Snowflake, Databricks



EDUCATION



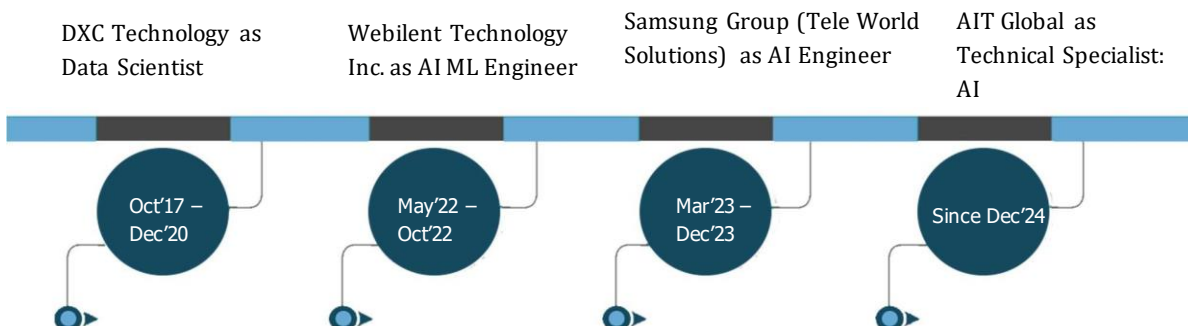
University of Houston, Houston, TX
Master of Science, Computer Science
Jan'21 – Dec'22



Medicaps Institute of Science & Technology
(Completed with Honors) Indore, MP
Bachelor of Engineering, Information Technology
Aug'13 – Jun'17



CAREER TIMELINE





WORK EXPERIENCE

Technical Specialist: AI | AIT Global | India

Dec'24 - Present

- ❖ Deployed the LangGraph-based multi-agent system locally with **Open-source LLMs (e.g., Mistral)** and validated its performance with custom test cases for knowledge retrieval and reasoning.
- ❖ Developed an NLP-based text similarity tool that extracts text from PDFs using Python's PDF readers, preprocesses the text using NLP techniques (tokenization, stopword removal, stemming), and computes similarity scores.
- ❖ Implemented Text-to-Speech (TTS) and Speech-to-Text (STT) systems using state-of-the-art models such as OpenAI Whisper and Google Cloud Speech API, enabling real-time speech synthesis and recognition for applications like virtual assistants, accessibility tools, and voice-based automation.
- ❖ Extracted text from resumes using PyMuPDF, or docx2txt, preprocessed and cleaned it with NLTK, leveraged LLMs (GPT 3.5, Claude Anthropic) or Hugging Face NER models (BERT) to extract key fields (Name, Contact, Skills, Experience, Education), and structured the data into JSON format.
- ❖ Conducted comprehensive analyses of user interactions and feedback to inform them about iterative improvements in AI applications, ensuring alignment with user needs and business objectives.

AI Engineer | Samsung Group (Tele World Solutions) | United States

Mar'23 - Dec'23

- ❖ Led the design and implementation of innovative GPT-based AI applications, utilizing advanced frameworks such as LangChain and LLaMA Index to create modular and scalable NLP pipelines tailored for diverse applications including content generation and chatbots.
- ❖ Enhanced the performance of pre-trained language models through fine-tuning techniques and hyperparameter optimization, achieving a notable improvement in model accuracy and efficiency.
- ❖ Engaged in collaborative efforts with cross-functional teams to deploy large language models (LLMs), ensuring the scalability of solutions capable of managing millions of monthly queries while maintaining high performance.
- ❖ Developed and maintained documentation for AI models and applications, facilitating knowledge transfer and ensuring compliance with best practices in software development and deployment.
- ❖ Provided mentorship and guidance to junior engineers, fostering a culture of continuous learning and innovation within the team.
- ❖ Conceptualized and developed a sophisticated Retrieval-Augmented Generation (RAG) system, improving information retrieval capabilities for chatbots by integrating external knowledge bases, thereby enhancing contextual accuracy by 35%.
- ❖ Achieved a 15% improvement in the performance of pre-trained language models through targeted fine-tuning and optimization efforts, showcasing a strong commitment to excellence in AI development.

AI ML Engineer | Webilent Technology Inc. | United States

May'22 - Oct'22

- ❖ Engineered a state-of-the-art chatbot utilizing advanced NLP techniques, focusing on enhancing user interaction and satisfaction through intelligent response generation and contextual understanding.
- ❖ Developed and implemented robust NLP pipelines for extracting valuable insights from unstructured data, improving processing efficiency and data utilization.
- ❖ Engaged in various NLP tasks, including Named Entity Recognition (NER), Part-of-Speech (POS) tagging, and machine translation, with a strong emphasis on optimizing model performance and accuracy.
- ❖ Collaborated with data scientists and engineers to refine machine learning models, ensuring alignment with project objectives and user requirements.
- ❖ Conducted thorough testing and validation of AI models, ensuring reliability and robustness in real-world applications.
- ❖ Participated in code reviews and knowledge-sharing sessions, contributing to the continuous improvement of development practices within the team.
- ❖ Successfully developed a chatbot that led to a 20% increase in customer satisfaction, showcasing the impact of innovative AI solutions on user experience.

Data Scientist | DXC Technology | India

Oct'17 - Dec'20

- ❖ Collaborated with a diverse team of over 12 professionals to integrate innovative ideas into product development, focusing on creating machine-learning solutions that address complex business challenges.
- ❖ Conducted comprehensive program analysis, implementing code modifications and updates to enhance the functionality and performance of existing applications.
- ❖ Managed data extraction, transformation, and loading (ETL) processes, ensuring data integrity and accuracy throughout the analytical workflow.
- ❖ Engaged in the implementation of suitable machine learning algorithms, tailoring solutions to meet specific project requirements and objectives.
- ❖ Successfully developed applications that reduced processing time by 70%, improving operational efficiency and productivity.



INTERNSHIP

Software developer | Zensar Technologies | India

Jan'17 - Jun'17

- ❖ Developed a comprehensive banking application using Java, focusing on the implementation of core banking functionalities and secure storage of customer details.
- ❖ Engaged in the design and development of user-friendly interfaces, ensuring a seamless experience for end-users while maintaining high standards of security and data integrity.



ACADEMIC PROJECTS

Authorship Attribution | Research Project NLP: Dataset preprocessing such as removal of the bullets and numbers, all the URLs from the text data, special characters and symbols, and text symbols that are not required. Also used stemming, tokenization, feature extraction & TF-IDF vectorizer. Calculated statistical inference from the data set including count, mean, median, max, and quartiles, and plotted various visualizations using TSNE projection, and matplotlib. Implemented supervised (topic modeling, K-means) and unsupervised methods (LR, NB, Random Forest, SVM, CNN).

Brain Science | R using Statistical Methods: Generating figures that illustrate the trends in cross-domain scholarship in Human Brain science using the R and Gephi tool. Evolution of Subject Areas boundary crossing within and across disciplinary clusters using Sankey diagrams. Calculating the logistics regression for the cross-domain.

Fake News Detection | Machine Learning: Classifying whether the given news is fake or real by implementing various classification methods compared to each other (Logistics regression, Naive Bayes, SVM, and Random Forest). Used FEVER data set which contains 185445 claims. Performance is improved using hyperparameter tuning and evaluated by calculating accuracy and F1 score.

Real-Time Sentiment Analysis Pipeline | Apache Spark & AWS: Developed a scalable real-time sentiment analysis pipeline using Apache Spark on AWS, processing large volumes of tweets/customer reviews. Implemented NLP-based sentiment classification using Spark MLlib and Hugging Face Transformers (BERT) for accurate text analysis. Built a cloud based big data architecture leveraging S3, Redshift and DynamoDB for efficient data processing. Deployed an interactive sentiment dashboard using Amazon QuickSight providing real-time insights on customer sentiment trends.